

Dengue fever is a human disease caused by the dengue virus.

Scientists designed an ELISA test to detect antibodies to the dengue virus in a patient's blood sample.

Figure 10 shows a diagram of this test and some information about how it works.

Figure 10

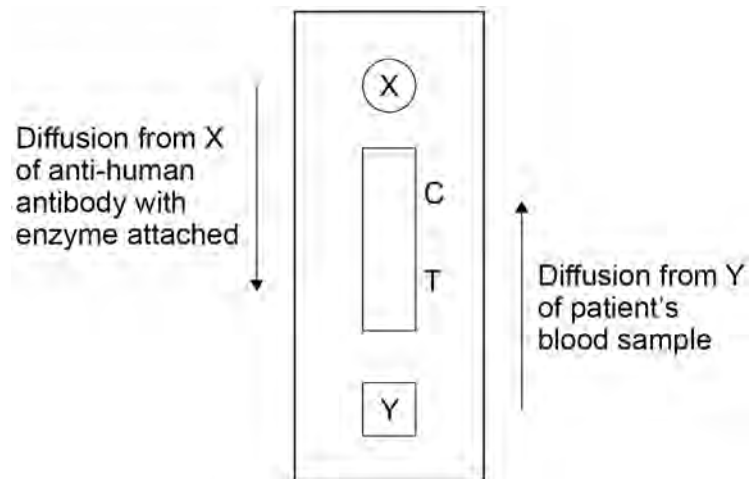
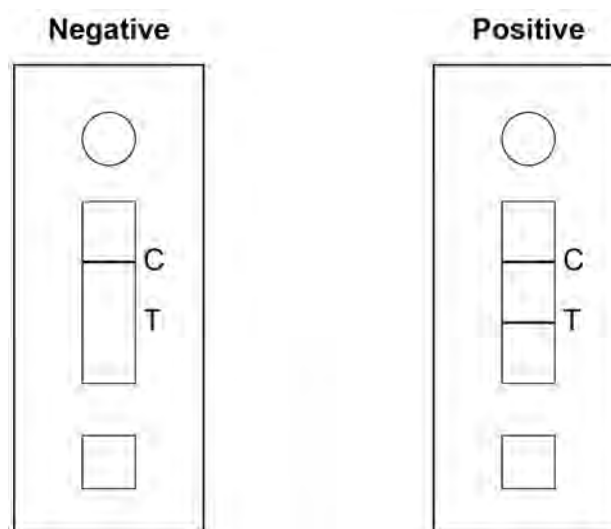


Figure 11 shows the negative and positive results that were produced 20 minutes after the use of the test shown in **Figure 10**.

Figure 11



0 9 . 1

Suggest what is on the test at line **T** and explain what causes the line to appear in a positive test.

[2 marks]

0 9 . 2

A line at **C** shows that the test has worked.

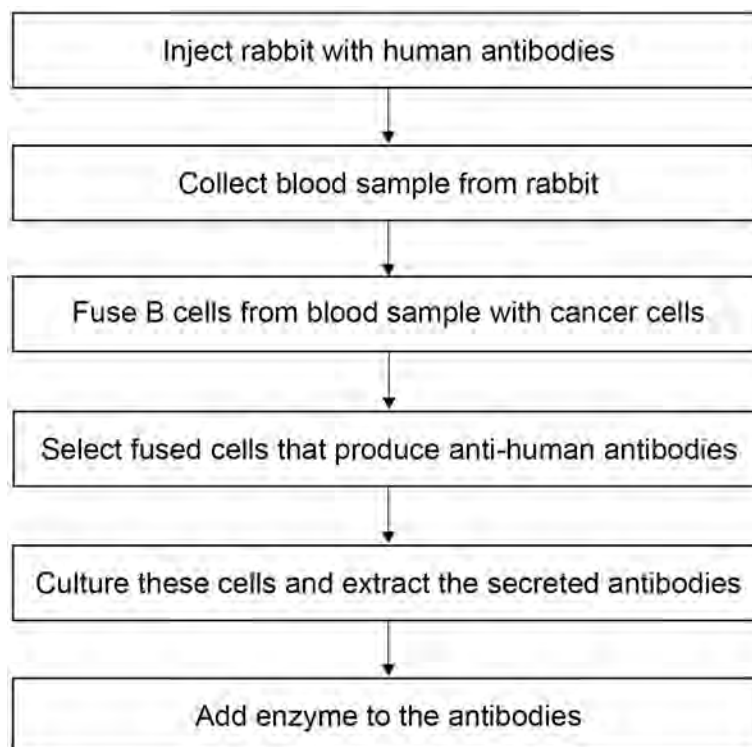
Suggest **one** reason why a line at **C** shows the test has worked.

[1 mark]

Question 9 continues on the next page

Figure 12 shows a flowchart of how the anti-human antibodies with enzyme attached are produced.

Figure 12



0 9 . 3 Suggest why the fused cells allow continuous production of monoclonal antibodies.

[2 marks]

0 9 . 4 Evaluate the ethics of the production process shown in **Figure 12**.

[1 mark]

Question 9 continues on the next page

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Early identification of dengue fever can be difficult as many other diseases produce the same symptoms. Early identification is important because people suffering with dengue fever can become ill very quickly and may need hospital treatment.

Scientists compared the effectiveness of three diagnostic tests for dengue fever.

- Laboratory-based test – a patient's blood sample is sent from the doctor's clinic to a laboratory for testing.
- Current test used in the doctor's clinic.
- New test to be used in the doctor's clinic – the ELISA test shown in **Figures 10** and **11** (on page 24).

The scientists' results are shown in **Table 3**.

A blood sample from each patient with confirmed dengue fever at each time after onset of symptoms was tested with all three diagnostic tests.

Table 3

Time after onset of symptoms / days	Number of confirmed dengue fever patients tested	Number of positive results		
		Laboratory-based test	Current test	New test
1–2	14	10	0	6
3–4	38	28	6	24
5–7	18	8	14	14

0 9 . 5

The scientists recommend that the new test is used for the identification of dengue fever in all countries around the world.

Discuss this recommendation. Use all the information given.

[3 marks]

0 9 . 6

The dengue virus causes damage to capillaries so that blood proteins move out of the capillaries into the tissue fluid.

Explain how this would affect the return of tissue fluid into the capillaries.

[2 marks]

1

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